

SCIENCE ONE, MATHEMATICS - HOMEWORK #2.3

Due on Tuesday, March 31st by 11PM.

1. Recall that the probability density function for the electron-proton distance r in the hydrogen atom ground state (i.e., the 1s orbital) is given by

$$\begin{aligned}f(r) &= 4\pi r^2 \psi_{100}^2(r) \\ &= Ar^2 e^{-\frac{2r}{a_0}},\end{aligned}$$

where a_0 is the Bohr radius.

- (a) Find the cumulative distribution function $F(r)$.
 - (b) Determine the value of A (i.e., “normalize” the distribution).
 - (c) Determine the probability that the electron will be found within $2a_0$ of the proton. Does this seem reasonable?
 - (d) Determine the “most probable” and the “expected” radius for the hydrogen atom in the ground state.
2. *Newton’s universal law of gravitation* states that the force of attraction between two point masses m and M has magnitude

$$F = \frac{GmM}{r^2},$$

where r is the distance between the masses and $G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$ is a constant.

- (a) If M represents the mass of the centre of the earth and we regard it as a point mass concentrated at its centre, show that Newton’s universal law of gravitation at the earth’s surface reduces to $F = mg$, where $g = 9.82 \text{ m/s}^2$. Assume for the calculation of M that the earth is a sphere with radius 6370 km and mean density $5.52 \times 10^3 \text{ kg/m}^3$.
- (b) Use the original $F = \frac{GmM}{r^2}$ with the earth regarded as a point mass to calculate the work required to lift a mass of 10kg from the earth’s surface to a height of 10 km.
- (c) Calculate the work in part (b) using the constant gravitational force $F = mg$ in part (a). Is there a significant difference?

3. In the next problem you will be working with a couple of examples of a fractal. A fractal is a mathematical set that displays a self-similarity property; that is, it exhibits a repeating pattern that displays at every scale.
- (a) The Cantor set, named after the German mathematician Georg Cantor (1845-1918), is constructed as follows. We start with the closed interval $[0, 1]$ and remove the open interval $(1/3, 2/3)$. That leaves the two intervals $[0, 1/3]$ and $[2/3, 1]$. We then remove the open middle interval third of each of these two remaining intervals. We continue this procedure indefinitely, at each step removing the open middle third of every interval that remains from the preceding step. The Cantor set consists of the numbers that remain in the original interval $[0, 1]$ after all those intervals have been removed.
- Sketch a diagram that shows the first 5 iterations of the Cantor set.
 - Give examples (at least 5) of some numbers in the Cantor set.
 - Show that the total length of all the intervals that are removed is 1. Despite that, the Cantor set is not an empty set.
- (b) The Sierpinski carpet is a two-dimensional counterpart of the Cantor set. It is constructed by removing the centre one-ninth of a square of side 1, then removing the centres of the eight smaller remaining squares, and so on. A visualization of the Sierpinski carpet is available on Wikipedia (https://en.wikipedia.org/wiki/Sierpinski_carpet).
- Show that the sum of the areas of the removed squares is 1. This implies that the Sierpinski carpet has area 0.